THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE UNIT TEST SET 2

Chapters 9–12: **Light - Reflection and Refraction, The Human Eye and the Colourful World, Electricity, Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Magnetic field lines inside a bar magnet go from 1

(a) the south pole to the north pole **(b)** the north pole to the south pole
(c) one side to the other randomly **(d)** do not exist

2. When light goes from air into glass, it 1

(a) gets reflected completely **(b)** bends towards the normal
(c) does not bend **(d)** bends away from the normal

3. Hypermetropia is corrected by using 1

(a) a prism **(b)** a convex lens **(c)** a concave mirror **(d)** a concave lens

4. The SI unit of resistance is 1

(a) watt **(b)** volt **(c)** ampere **(d)** ohm

5. A soft iron core inside a solenoid 1

(a) makes an electromagnet **(b)** reduces the field
(c) stops the current **(d)** has no effect

6. The power of a lens of focal length 50 cm is 1

(a) +5 D **(b)** +0.5 D **(c)** +2 D **(d)** -2 D

7. Which colour bends least on passing through a glass prism? 1

(a) Red **(b)** Violet **(c)** Green **(d)** Blue

8. In a series circuit the quantity that is the same through all the resistors is 1

(a) power **(b)** resistance
(c) current **(d)** potential difference

9. The phenomenon of electromagnetic induction was discovered by 1

(a) Michael Faraday **(b)** Ampere **(c)** Newton **(d)** Oersted

10. The centre of the reflecting surface of a spherical mirror is the 1

(a) centre of curvature **(b)** aperture
(c) principal focus **(d)** pole

11. The size of the pupil is controlled by the 1

(a) optic nerve **(b)** iris **(c)** retina **(d)** ciliary muscles

12. Electric current is measured by an 1

(a) voltmeter connected in series **(b)** ammeter connected in series
(c) voltmeter connected in parallel **(d)** ammeter connected in parallel

13. Two magnetic field lines 1

(a) intersect at the centre **(b)** intersect at the poles
(c) intersect often **(d)** never intersect each other

14. When light passes from a denser to a rarer medium, it 1

(a) stops **(b)** bends away from the normal
(c) bends towards the normal **(d)** goes straight

15. In myopia, the image of a distant object is formed 1

(a) on the iris **(b)** behind the retina
(c) on the retina **(d)** in front of the retina

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. What are the causes of myopia? 2

17. List the factors on which the resistance of a conductor depends. 2

18. Name the rule to find the direction of the force on a current-carrying conductor and state it. 2

19. What is the lateral displacement? 2

20. What is the Tyndall effect? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A proton enters a magnetic field perpendicular to it. State what happens and the direction of force. 3

22. What happens to the speed and wavelength of light when it passes from air to glass? 3

23. Why do planets not twinkle? 3

24. Why does a bulb filament glow but the connecting wire does not? 3

OR

What are the colour codes of the live, neutral and earth wires? 3

25. An object is placed at a distance of 30 cm in front of a concave mirror of focal length 15 cm. Find the position and nature of the image. 3

OR

How do the ciliary muscles help in focusing on near and far objects? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Define power of a lens and state its unit. A convex lens of focal length 25 cm and a concave lens of focal length 50 cm are in contact. Find the net power. 5

OR

What is dispersion of light? Describe Newton's experiment with two prisms. 5

27. Establish the formula H = I^2 R t from the definitions of work and potential difference. 5

OR

Draw (describe) a domestic electric circuit and explain the functions of the live wire, neutral wire, earth wire and fuse. 5

END OF QUESTION PAPER **Total: 50 Marks**

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